

Arpan Mondal

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PROFESSIONAL SUMMARY

Self-taught software engineer with hands-on experience building distributed backend systems, Linux-based workflows, concurrent services, data-intensive applications, and production-grade user interfaces. Strong foundation in algorithms and scalable system development with FastAPI, Node.js, Docker, Google Cloud, React, Flutter, and Redux Toolkit.

PROFESSIONAL EXPERIENCE

Software Engineer - MealOBox, Remote (Dec 2023 - Present)

- Refactored Flutter UI architecture, improving rendering efficiency by 40% and making screens easier to scale across product modules.
- Built real-time interactive mobile modules that reduced UI latency by 30% and improved responsiveness for user-facing workflows.
- Worked directly under the CEO/founder's guidance during early mobile development, translating Figma UI/UX designs into Flutter interfaces and production-ready frontend flows.
- Developed admin dashboard web modules with real data fetching, API integration, and Redux Toolkit state management for operational views.
- Contributed to backend API flows using async pipelines while collaborating in Agile teams on performance, testing, and scalable modules.

PROJECTS

Raw-Gent - Distributed AI-Powered Code Assistant | [GitHub](#) | [Live](#)

Tech: FastAPI, Python, Docker, Google Cloud Run, React, Google ADK

- Designed a distributed FastAPI backend with async workers for agentic code workflows, improving throughput for long-running analysis and generation tasks.
- Implemented GitHub OAuth, automated testing, and pull request generation pipeline for AI-assisted debugging, review, and feature creation.
- Built multimodal Google ADK agent flows to support repository understanding, code review, issue investigation, and implementation workflows.

Help-Now-India - Real-Time Emergency System | [GitHub](#)

Tech: Node.js, Express, WebSockets, React Native

- Engineered an event-driven backend supporting sub-120ms emergency alert delivery through real-time WebSocket communication.
- Implemented geolocation routing and proximity-based dispatch using spatial queries to connect emergency requests with nearby responders.

StylerOP - AI Image Transformation Platform | [GitHub](#) | [Live](#)

Tech: FastAPI, Python, MongoDB, Google Gemini, React, TypeScript, Redux Toolkit, Three.js

- Built a full-stack AI image transformation app with upload preview, style selection, transformation history, and downloadable generated outputs.
- Implemented a FastAPI image pipeline that validates file type and size, processes uploads with PIL, and returns base64 PNG results generated through Gemini.
- Added JWT/cookie authentication, MongoDB-backed refresh token storage, Redux Toolkit async state handling, protected routes, and typed API validation.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, Dart

Systems: Distributed Systems, Linux/Unix, TCP/IP, Concurrency, Microservices, Event-driven Systems, Async Pipelines

Frameworks: FastAPI, Node.js, Express, React, Flutter, Django

Databases: PostgreSQL, MongoDB

Cloud and Tools: Docker, Google Cloud Run, CI/CD, WebSockets, Git, REST APIs

AI/ML: Google ADK, LLM Integration, Multimodal Agent Workflows, Google Gemini